mhs-poisson-ltau: implementation

Technical notes on `run.mode = "mhs-poisson-ltau"`. HSEMHS, 26 Oct 2026 (after the embedding-API merge, 9d89191).

1. Running it

```
[mpiexec -n R] julia --project=. -t T scripts/run.jl config/modes/mhs-poisson-ltau.toml
```

Driver `run_mhs_poisson_ltau` (`src/RunModes.jl`): read, `prepare_embedded(COMM_WORLD, :mhs_poisson_ltau, ...)`, `solve_embedded!`, write, `free_embedded!` (`src/Embedded.jl`); solver `PoissonTau.solve_tau` (`src/PoissonTau.jl`); MPI layer `src/Slab.jl`; PCG and threaded vector updates from `PoissonZ`. Input and `[tau_grid]` keys as `column-hse-ltau`; in addition:

Key	Default	Meaning
`poisson.logtau_cut`	-4	solved levels $\log\tau \geq$ cut (nearest level k_c)
`poisson.max_steps`	100	Levenberg–Marquardt steps
`poisson.rtol`, `n_consec`	10^{-4}, 3	stop when $(\Phi_{\rm prev}-\Phi)/\Phi_{\rm prev} <$ rtol on `n_consec` consecutive accepted steps (objective only)
`poisson.cg_tol`, `cg_maxit`	10^{-3}, 80	PCG per trial
`poisson.dc_cap_m`	3×10^5	largest height change of a column per step (whole-step scaling)
`poisson.lm_mu0`	0	initial damping

2. Flow

`column_hse_tau` (aligned, gauged; the current column integrator without a cap on the rise of $\ln P$ and with the end-slope substep rule, see the `column-hse-ltau` notes) gives $u=\ln P$ and z_0 on all levels. If any column fails there (pressure outside the table), the solve is not attempted and the driver raises an error. `solve_tau` solves for u on levels $k>k_c$ and $c=z_{k_c}$ per column, with $u_{k\le k_c}$ and $z_k-z_{k_c}$ $(k<k_c)$ fixed; a start state outside the table returns `reason = :eos_out_of_envelope` (driver error) instead of throwing. Then `gauge!` (mean $z(\tau{=}1)=0$), `tau_state!` (ρ, n_e, χ per node), `check_fields!` (as in `column-hse-ltau`), output in the `column-hse-ltau` layout.

Option checks in `prepare_embedded`, identical on all ranks: `logtau_cut` inside the axis; the cut level $k_c\geq 2$ (the height stencil at k_c uses level k_c-1); at least 3 solved levels; `max_steps`, `n_consec`, `cg_maxit` ≥ 1; `rtol` >0; $0<$ `cg_tol` <1; `dc_cap_m` >0; `lm_mu0` ≥ 0. The distributed FFT plan (`SlabFFT`) is built once in `prepare_embedded` and reused by every solve.

3. State, heights and residuals (`evaluate!`)

Nodes $m=1\ldots n_1$ are levels $k=k_c\ldots n_\tau$. The unknown vector per column has n_1 slots: slot $1=c$, slots $2\ldots n_1=u$ at $k>k_c$ (one array (n_1,N_y,n_x), so FFTs and halos treat it uniformly). At each node the EOS gives $\ln\rho$, $\partial\ln\rho/\partial\ln P$ and $\ln\chi$; $\partial\ln\chi/\partial\ln P$ by a central difference ($\varepsilon=10^{-4}$, the table returns no opacity derivative). With $a=\tau/\chi$, $q=\rho/P$:

$$z_{k+1}=z_k-\tfrac12\Delta\ln\tau_k\,(a_k+a_{k+1}),\qquad z_{k_c}=c.$$

Lorentz force at all levels (`lorentz_tau!`): for each component f of $\mathbf{B}$, $\partial_z f=(f_{k-1}-f_{k+1})/(z_{k-1}-z_{k+1})$ (one-sided at the ends), $\partial_x f|_z=D_x f-D_x z\,\partial_z f$, likewise y; $\mathbf{J}=\nabla\times\mathbf{B}/\mu_0$; $\mathbf{L}=\mathbf{J}\times\mathbf{B}$. Residuals with $W=gq-L_z e^{-u}$, $w=Wa$:

$$R_x=-D_x u-W\,D_x z+L_x e^{-u},\qquad R_{z,f}=u_{f+1}-u_f-\tfrac12\Delta\ln\tau_f\,(w_f+w_{f+1}).$$

Weights from the start state (fixed): $\bar H_m$ global median of $1/(gq)$, $\overline{\Delta z}_f$ global median of the level thickness, $w_c=\overline{\Delta z}_{\rm node}\bar H^2$, $w_f=\bar H_f\bar H_{f+1}/\overline{\Delta z}_f$; $\Phi=\sum w_c(R_x^2+R_y^2)+\sum w_f R_z^2$.

4. Jacobian

Exact, including $\mathbf{L}$'s dependence on the heights (`Jop!`, adjoint `JTop!`):

- **Triangular operator.** $\delta z_m=\delta c+\sum_{f<m}\tfrac12\Delta\ln\tau_f\,(a\chi_u)_f\delta u_f+(a\chi_u)_{f+1}\delta u_{f+1}$ with $\chi_u=\partial\ln\chi/\partial\ln P$: a cumulative sum down each column. Its adjoint is a reverse cumulative sum.
- Vertical: $\delta R_{z,f}=\delta u_{f+1}-\delta u_f-\tfrac12\Delta\ln\tau_f(w'_f\delta u_f+w'_{f+1}\delta u_{f+1})$, $w'=\partial w/\partial u=W'a-Wa\chi_u$, $W'=gq(\rho_u-1)+L_z e^{-u}$.
- Horizontal: $\delta R_x=-D_x\delta u-W\,D_x\delta z+\ell_x\delta u$, $\ell_x=-D_x z\,W'-L_x e^{-u}$.
- **Lorentz through the heights.** With $\zeta_m=(\delta z_{m-1}-\delta z_{m+1})/(z_{m-1}-z_{m+1})$ (the relative change of the vertical spacing; δz above the cut $=\delta c$): $\delta(\partial_z f)=-\partial_z f\,\zeta$, $\delta(\partial_x f|_z)=-\partial_z f\,D_x\delta z+D_x z\,\partial_z f\,\zeta$. Hence $\delta\mathbf{J}=$

$\mathbf{J}_\zeta\zeta+\mathbf{J}_X D_x\delta z+\mathbf{J}_Y D_y\delta z$ with $\mathbf{J}_\zeta = (z_y B_{z,z}+B_{y,z},\ -B_{x,z}-z_x B_{z,z},\ z_x B_{y,z}-z_y B_{x,z})/\mu_0$, $\mathbf{J}_X = (0, B_{z,z}, -B_{y,z})/\mu_0$, $\mathbf{J}_Y = (-B_{z,z}, 0, B_{x,z})/\mu_0$, and $\delta\mathbf{L} = \delta\mathbf{J}\times\mathbf{B}$. It enters δR_x through $e^{-u}(\delta L_x + z_x\delta L_z)$ and δR_z through $e^{-u}a\,\delta L_z$: nine node-local coefficient arrays.

- Checks (`MHS_CHECK` in the script version): adjoint $\langle Jv, r\rangle = \langle v, J^\top r\rangle$ to 4×10^{-16}; Jv against central differences of $\mathbf{R}$ to 10^{-9} (with $\mathbf{B}$), 10^{-10} (without). With $\mathbf{L}$ held fixed the directional error was 100 % in magnetic regions and the solve stalled after one step.

5. Linear solve and preconditioner

Normal equations $(J^\top W J + E + \mu D)\,\delta = -J^\top W\mathbf{R}$ by PCG with global inner products; $E = 10^{-2}\kappa^2 D$, $D = (\sum w_c\bar{W}^2,\ w_{c,2},\dots)$ (the damping shape; slot 1 for c, which no vertical equation sees).

Preconditioner (`setprec!`, `Minv!`): the operator with layer-median coefficients ($\overline{a\chi_u}$, $\overline{w'}$, $\bar{W}$) is, per horizontal mode, $M(s^2) = A + s^2 B$ with s^2 as in `mhs-poisson-z`; $A = G_z^\top W_f G_z + E + \mu D$ (vertical rows), $B = G_h^\top W_c G_h$, $G_h = -(I + \bar{W}\,S)$ the horizontal symbol divided by ik including the triangular matrix S. Both are dense $n_1\times n_1$ ($n_1 = 55$). Once per step: Cholesky $A = LL^\top$, eigen-decomposition $L^{-1}BL^{-\top} = V\Lambda V^\top$, $Q = V^\top L^{-1}$; then $M(s^2)^{-1} = Q^\top\,\mathrm{diag}(1/(1+s^2\lambda))\,Q$ for every mode: two complex GEMMs over all local modes of the transposed FFT layout (`SlabFFT`, as in `mhs-poisson-z`), BLAS threaded during the solve.

6. Steps

Levenberg–Marquardt: for μ (start `lm_mu0`) solve, scale the *whole* step by $\min(1,\ 2/\max|\delta u|,\ \texttt{dc_cap}/\max|\delta c|)$, try up to 4 halvings on the true Φ (EOS status checked); if none decreases Φ, $\mu \leftarrow \max(4\mu, 1)$ and re-solve (up to 8 times). After acceptance (source rule): $\mu \leftarrow \mu/3$ if the accepted step length factor is exactly $\alpha = 1$; unchanged if $\alpha \ge s_c/2$ (s_c the whole-step scale); $\times2$ otherwise; $\mu < 10^{-3} \to 0$. When the whole-step scaling is active ($s_c < 1$), a full scaled step ($\alpha = s_c$) therefore does *not* lower μ; whether that was intended is a code question (follow-up). Element-wise clamping of δc (the first version) turned the direction into an ascent direction where the step asked for > 1 Mm in a few columns: Φ rose linearly with the step length.

Termination. (a) $(\Phi_{\mathrm{prev}} - \Phi)/\Phi_{\mathrm{prev}} <$ `rtol` on `n_consec` consecutive accepted steps (log line `converged: ...`); (b) no damping level gives a decrease (`no decrease up to μ = ...; stopping`); (c) `max_steps` reached (no message). The reason (`:converged`, `:no_descent`, `:max_steps`) is returned in the status and written as the output attribute `stop_reason`. Stopping uses the objective only. Two earlier rules are gone: a single small step (it stopped one run after 17 steps while strong-field columns still moved 80 km per step), and a height shortcut (one small step plus $\max|\delta c| < 100$ m), removed on 24 Oct 2026 in line with the Φ-only policy. The logged $\max|\delta c|$ is the largest change of a column's *cut-level* height in the step, not the change of its $\tau = 1$ height (which also depends on the changed layer thicknesses). The file driver still returns `converged`, exit 0, in all three cases: read `stop_reason` or the log (code follow-up: propagate the termination into the exit status). A met rule does not certify small residuals.

Stationarity diagnostic. After the iteration, at the returned state and with $\mu = 0$, one more system $(J^\top W J + E)\,\delta = -J^\top W\mathbf{R}$ is solved by PCG (diagnostic only; u, z unchanged); logged as `stationarity: Gauss-Newton decrement ...` and written as `gn_decrement` $= \delta\cdot(-J^\top W\mathbf{R})/\Phi$, the relative decrease a further undamped step would predict (0 at a stationary point); $\|J^\top W\mathbf{R}\|$ and the final μ are in the embedded status. Option `rms_atol` (embedded API only, default 0) accepts a start with $\sqrt{\Phi_0/W} \le$ `rms_atol` in zero steps.

7. Parallelisation

x slabs; halo exchange in x for every D_x (heights, u, $\mathbf{B}$, and in the Jacobian); global sums; exact global medians for the weights and the preconditioner (bisection on order-preserving integer keys); the distributed FFT (rfft in y, `Alltoallv` transpose, FFT in x) and GEMMs on the local k_y slab. Threads in all node loops; the standalone driver raises BLAS to `min(32, nthreads)` around the solve and resets it to 1 (the embedded API leaves BLAS to the caller); FFTW plans are single-threaded, applied from Julia threads. The solve is *step-path sensitive*: on a 24×18 crop the same one-rank solve changes by O(1) in a few strong-field column heights between 1 and 4 BLAS threads (different accepted LM steps), while Φ/Φ_0 agrees (8.473×10^{-3} vs 8.474×10^{-3}).

The field values on the τ levels are used unchanged at the new heights; no discrete $\nabla\cdot\mathbf{B} = 0$ is imposed (the staggered construction of `full-mhs-z` needs a fixed geometry). Above the cut the levels stay at their column-HSE pressures and relative heights and move rigidly with c; the global height offset is fixed only by the final gauge.

8. Verification and performance

- Automated tests: `test_embedded.jl` (stored standalone outputs on two windows of the production input, 24×18 and 11×12 columns, reproduced to $\le 10^{-12}$; reason in {converged, max_steps, no_descent}; finite, non-negative GN decrement; a balanced start accepted in zero steps; 1 vs 30 steps gives a larger decrement after 1); `mpi/mpi_embedded.jl`: on the converged 11×12 window (5 steps) 2, 3, 4 ranks agree with one rank to

$\leq 9\times10^{-11}$ (asserted 10^{-9}); on the 24×18 window (24 steps) the differences are O(1) in a few heights (step-path sensitivity, above) and only recorded.

- 96^2 magnetic crop (manual), ranks × threads 1×8, 1×32, 4×8, consecutive-step rule: 37 steps each; $z(\tau{=}1)$ differences median 1 m, 99th percentile 12–24 m, max 150 m (weakly determined strong-field columns). Agreement to rounding only; identical trajectories across layouts are not promised. After the removal of the height shortcut (commit 813a3af) the 1×32 run again took 37 steps.
- Against the original script `scripts/poisson/poisson_tau_direct.jl` (single-step rule) on the same crop: $z(\tau{=}1)$ differences median 3.6 m, 99th percentile 95 m, max 1 km, because the two stopped one step apart while some columns were still moving.
- Full cube ($504\times504\times95$, 54 solved $\ln P$ levels + 1 height per column, 1.4×10^7 unknowns), with the superseded single-step rule: one node, 96 threads, 17 steps, 14 min, 11 GB (commit b96bb4e); 8 ranks on 4 nodes, 8.3 min, 3.1 GB per rank, same summary statistics (commit 13666b7, log `out/modes/mhs-poisson-ltau.log` in the cluster worktree `MHS_modes`). These runs predate the removal of the $\ln P$ cap in the starting state; the current code has not been run on the full cube. PCG reaches its 80-iteration cap in every step (relative residual 10^{-2}–2×10^{-1}): the next speed-up is a better preconditioner.

9. Result on the production cube

$z(\tau{=}1)$ relative to the cube mean, median [5 %, 95 %] in km, by $|\mathbf{B}|$ at $\tau = 1$ (fixed-point = the superseded scheme; 24 Oct 2026, starting state still with the $\ln P$ cap, single-step stop rule):

$\lvert\mathbf{B}\rvert$ at $\tau = 1$	column HSE in τ	fixed-point scheme	**mhs-poisson-ltau**
0–200 G	−38 [−85, 187]	−55 [−257, 380]	−10 [−35, 73]
200–500 G	−68 [−88, 58]	+120 [−180, 549]	−26 [−52, 24]
500–1000 G	−86 [−104, −36]	+183 [−198, 587]	−51 [−89, −17]
1000–1500 G	−121 [−144, −92]	+314 [−117, 563]	−96 [−125, −63]
1500–3000 G	−156 [−175, −140]	+402 [58, 568]	−130 [−167, −104]

Force imbalance after the solve, RMS in units of ρg: 0.04–0.09 horizontal and 0.04–0.06 vertical for $\log\tau < 0$ (start 0.3–0.8).

10. Design history

1. Fixed point (`scripts/poisson/poisson_tau.jl`): map to a height grid, Poisson solve there, re-integrate the τ columns, repeat. Each pass is a full Poisson solve (1 h on the full cube), and it diverged where $\mathbf{L}$ carries the gas: the heights were integrated with a pressure that was then replaced. A corrected variant (heights integrated with the Poisson pressure) settled but lifted the strong-field columns instead of depressing them (median +400 km above the cube mean for 1500–3000 G).
2. Direct solve with $\mathbf{L}$ fixed in the Jacobian: exact without $\mathbf{B}$, stalled in magnetic regions.
3. Exact Jacobian + LM with whole-step scaling + the consecutive-step stop rule: the present mode.